

SQL / DBMS Short Notes

1. Basic Queries

SELECT: Retrieve data from table.

WHERE: Filter records based on condition.

ORDER BY: Sort results ascending/descending.

LIMIT: Restrict number of rows.

2. Joins

INNER JOIN: Matching records from both tables.

LEFT JOIN: All from left + matched from right.

RIGHT JOIN: All from right + matched from left.

FULL JOIN: All records from both tables.

3. Aggregate Functions

COUNT(): Number of rows.

SUM(): Total value.

AVG(): Average value.

MAX()/MIN(): Highest/lowest value.

4. GROUP BY & HAVING

GROUP BY: Group rows for aggregation.

HAVING: Filter grouped data.

5. Keys

Primary Key: Unique identifier.

Foreign Key: Reference to another table.

6. Normalization

1NF: Remove repeating groups.

2NF: Remove partial dependency.

3NF: Remove transitive dependency.

7. Transactions (ACID)

Atomicity: All or nothing.

Consistency: Valid state.

Isolation: Independent execution.

Durability: Permanent changes.

8. Indexing

Improves query speed by indexing columns.